

Implementasi Structural Equation Modeling Partial Least Square (SEM-PLS) pada Customer Behavior E-Commerce

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Pendahuluan

Penelitian ini bertujuan untuk menganalisis pengaruh Membership Type dan Discount Applied terhadap Customer Satisfaction dengan Average Rating sebagai variabel mediasi pada platform e-commerce menggunakan metode SEM-PLS.

Metode yang digunakan adalah Structural Equation Modeling - Partial Least Squares (SEM-PLS).

Load Library

```
library(seminr)
library(readxl)
```

Import Dataset

```
data <- read.csv("E-commerce Customer Behavior - Sheet1.csv")
```

Struktur Dataset

```
names(data)
```

```
## [1] "Customer.ID"      "Gender"
## [3] "Age"              "City"
## [5] "Membership.Type"   "Total.Spend"
## [7] "Items.Purchased"   "Average.Rating"
## [9] "Discount.Applied"  "Days.Since.Last.Purchase"
## [11] "Satisfaction.Level"
```

```
str(data)
```

```
## 'data.frame': 350 obs. of 11 variables:
## $ Customer.ID : int 101 102 103 104 105 106 107 108 109 110 ...
## $ Gender : chr "Female" "Male" "Female" "Male" ...
## $ Age : int 29 34 43 30 27 37 31 35 41 28 ...
## $ City : chr "New York" "Los Angeles" "Chicago" "San Francisco" ...
## $ Membership.Type : chr "Gold" "Silver" "Bronze" "Gold" ...
## $ Total.Spend : num 1120 780 511 1480 720 ...
## $ Items.Purchased : int 14 11 9 19 13 8 15 12 10 21 ...
## $ Average.Rating : num 4.6 4.1 3.4 4.7 4 3.1 4.5 4.2 3.6 4.8 ...
## $ Discount.Applied : logi TRUE FALSE TRUE FALSE TRUE FALSE ...
## $ Days.Since.Last.Purchase: int 25 18 42 12 55 22 28 14 40 9 ...
## $ Satisfaction.Level : chr "Satisfied" "Neutral" "Unsatisfied" "Satisfied" ...
```

```
head(data)
```

```
## Customer.ID Gender Age City Membership.Type Total.Spend
## 1 101 Female 29 New York Gold 1120.20
## 2 102 Male 34 Los Angeles Silver 780.50
## 3 103 Female 43 Chicago Bronze 510.75
## 4 104 Male 30 San Francisco Gold 1480.30
## 5 105 Male 27 Miami Silver 720.40
## 6 106 Female 37 Houston Bronze 440.80
## Items.Purchased Average.Rating Discount.Applied Days.Since.Last.Purchase
## 1 14 4.6 TRUE 25
## 2 11 4.1 FALSE 18
## 3 9 3.4 TRUE 42
## 4 19 4.7 FALSE 12
## 5 13 4.0 TRUE 55
## 6 8 3.1 FALSE 22
## Satisfaction.Level
## 1 Satisfied
## 2 Neutral
## 3 Unsatisfied
## 4 Satisfied
## 5 Unsatisfied
## 6 Neutral
```

Variabel yang digunakan

```
data_sem <- data[, c(
  "Membership.Type",
  "Discount.Applied",
  "Average.Rating",
  "Satisfaction.Level"
)]
```

Melihat Isi Data

```
unique(data_sem$Membership.Type)
```

```
## [1] "Gold" "Silver" "Bronze"
```

```
unique(data_sem$Satisfaction.Level)
```

```
## [1] "Satisfied" "Neutral" "Unsatisfied" ""
```

Preprocessing

Mengubah Membership Type Menjadi Numerik

```
data_sem$Membership.Type <- as.numeric(factor(data_sem$Membership.Type, levels = c("Bronze", "Silver", "Gold")))
```

Mengubah Satisfaction Level Menjadi Numerik

```
data_sem$Satisfaction.Level <- as.numeric(factor(data_sem$Satisfaction.Level, levels = c("Unsatisfied", "Neutral", "Satisfied")))
```

Mengubah Discount Applied Menjadi Numerik

```
data_sem$Discount.Applied <- ifelse(data_sem$Discount.Applied == TRUE, 1, 0)
```

Menghapus Missing Value

```
data_sem <- na.omit(data_sem)
```

Cek Struktur data

```
str(data_sem)
```

```
## 'data.frame': 348 obs. of 4 variables:
## $ Membership.Type : num 3 2 1 3 2 1 3 2 1 3 ...
## $ Discount.Applied : num 1 0 1 0 1 0 1 0 1 0 ...
## $ Average.Rating : num 4.6 4.1 3.4 4.7 4 3.1 4.5 4.2 3.6 4.8 ...
## $ Satisfaction.Level: num 3 2 1 3 1 2 3 2 1 3 ...
## - attr(*, "na.action")= 'omit' Named int [1:2] 72 144
## .. attr(*, "names")= chr [1:2] "72" "144"
```

Measurement Model

```
measurement_model <- constructs(  
  
  composite(  
    "Membership",  
    single_item("Membership.Type")  
  ),  
  
  composite(  
    "Discount",  
    single_item("Discount.Applied")  
  ),  
  
  composite(  
    "Rating",  
    single_item("Average.Rating")  
  ),  
  
  composite(  
    "Satisfaction",  
    single_item("Satisfaction.Level")  
  )  
  
)
```

Structural Model

```
structural_model <- relationships(  
  
  paths(  
    from = c("Membership", "Discount"),  
    to = c("Rating", "Satisfaction")  
  ),  
  
  paths(  
    from = "Rating",  
    to = "Satisfaction"  
  )  
  
)
```

Estimasi SEM-PLS

```
pls_model <- estimate_pls(  
  data = data_sem,  
  measurement_model = measurement_model,  
  structural_model = structural_model  
)
```

```
## Generating the semnr model
```

```
## All 348 observations are valid.
```

Hasil SEM-PLS

```
summary(pls_model)
```

```
##  
## Results from package semnr (2.5.0)  
##  
## Path Coefficients:  
##           Rating Satisfaction  
## R^2       0.908       0.747  
## AdjR^2    0.907       0.745  
## Membership 0.950       1.160  
## Discount  -0.073      -0.455  
## Rating      .       -0.442  
##  
## Reliability:  
##           alpha rhoA rhoC AVE  
## Membership  1.000 1.000 1.000 1.000  
## Discount    1.000 1.000 1.000 1.000  
## Rating      1.000 1.000 1.000 1.000  
## Satisfaction 1.000 1.000 1.000 1.000  
##  
## Alpha, rhoA, and rhoC should exceed 0.7 while AVE should exceed 0.5
```

Bootstrapping

```
boot_model <- bootstrap_model(  
  semnr_model = pls_model,  
  nboot = 5000,  
  cores = 1  
)
```

```
## Bootstrapping model using semnr...
```

```
## SEminR Model successfully bootstrapped
```

Hasil Bootstrapping

```
summary(boot_model)
```

```

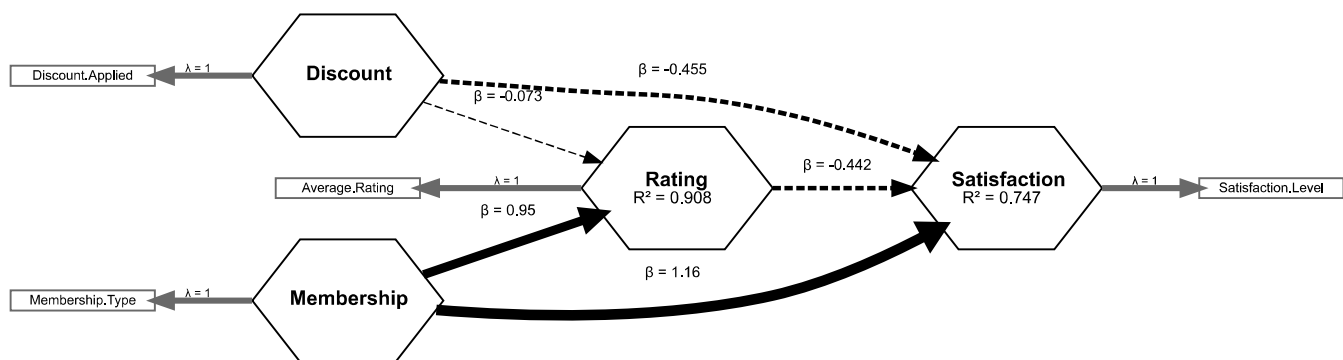
##
## Results from Bootstrap resamples: 5000
##
## Bootstrapped Structural Paths:
##
## Original Est. Bootstrap Mean Bootstrap SD T Stat.
## Membership -> Rating 0.950 0.950 0.005 191.284
## Membership -> Satisfaction 1.160 1.160 0.094 12.356
## Discount -> Rating -0.073 -0.073 0.017 -4.308
## Discount -> Satisfaction -0.455 -0.456 0.027 -16.924
## Rating -> Satisfaction -0.442 -0.442 0.084 -5.234
##
## 2.5% CI 97.5% CI Bootstrap P Val
## Membership -> Rating 0.940 0.959 0.000
## Membership -> Satisfaction 0.970 1.341 0.000
## Discount -> Rating -0.107 -0.040 0.000
## Discount -> Satisfaction -0.508 -0.404 0.000
## Rating -> Satisfaction -0.605 -0.276 0.000
##
## Bootstrapped Weights:
##
## Original Est. Bootstrap Mean Bootstrap SD
## Membership.Type -> Membership 1.000 1.000 0.000
## Discount.Applied -> Discount 1.000 1.000 0.000
## Average.Rating -> Rating 1.000 1.000 0.000
## Satisfaction.Level -> Satisfaction 1.000 1.000 0.000
##
## T Stat. 2.5% CI 97.5% CI Bootstrap P Val
## Membership.Type -> Membership . 1.000 1.000 0.000
## Discount.Applied -> Discount . 1.000 1.000 0.000
## Average.Rating -> Rating . 1.000 1.000 0.000
## Satisfaction.Level -> Satisfaction . 1.000 1.000 0.000
##
## Bootstrapped Loadings:
##
## Original Est. Bootstrap Mean Bootstrap SD
## Membership.Type -> Membership 1.000 1.000 0.000
## Discount.Applied -> Discount 1.000 1.000 0.000
## Average.Rating -> Rating 1.000 1.000 0.000
## Satisfaction.Level -> Satisfaction 1.000 1.000 0.000
##
## T Stat. 2.5% CI 97.5% CI Bootstrap P Val
## Membership.Type -> Membership . 1.000 1.000 0.000
## Discount.Applied -> Discount . 1.000 1.000 0.000
## Average.Rating -> Rating . 1.000 1.000 0.000
## Satisfaction.Level -> Satisfaction . 1.000 1.000 0.000
##
## Bootstrapped HTMT:
##
## Original Est. Bootstrap Mean Bootstrap SD 2.5% CI
## Membership -> Discount 0.004 0.043 0.032 0.002
## Membership -> Rating 0.950 0.950 0.003 0.943
## Membership -> Satisfaction 0.742 0.741 0.022 0.697
## Discount -> Rating 0.077 0.080 0.048 0.005
## Discount -> Satisfaction 0.425 0.426 0.045 0.335
## Rating -> Satisfaction 0.695 0.694 0.024 0.647
##
## 97.5% CI Bootstrap P Val
## Membership -> Discount 0.120 0.000
## Membership -> Rating 0.956 0.000

```

```
## Membership -> Satisfaction    0.784        0.000
## Discount -> Rating           0.184        0.000
## Discount -> Satisfaction     0.513        0.000
## Rating -> Satisfaction        0.740        0.000
##
## Bootstrapped Total Paths:
##                               Original Est. Bootstrap Mean Bootstrap SD 2.5% CI
## Membership -> Rating          0.950          0.950          0.005   0.940
## Membership -> Satisfaction    0.740          0.740          0.024   0.692
## Discount -> Rating           -0.073         -0.073          0.017  -0.107
## Discount -> Satisfaction     -0.422         -0.424          0.028  -0.479
## Rating -> Satisfaction        -0.442         -0.442          0.084  -0.605
##
##                               97.5% CI
## Membership -> Rating          0.959
## Membership -> Satisfaction    0.785
## Discount -> Rating           -0.040
## Discount -> Satisfaction     -0.368
## Rating -> Satisfaction        -0.276
```

Visualisasi Model

```
plot(pls_model)
```



Kesimpulan

Berdasarkan hasil analisis SEM-PLS, penelitian ini dapat digunakan untuk mengetahui pengaruh Membership Type dan Discount Applied terhadap Customer Satisfaction melalui Average Rating pada platform e-commerce.